Roshan Y Singh

cs23btech11052@iith.ac.in | roshanroshys@gmail.com | +91 8197079584 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.80
Oxford Independent PU College, Bangalore	Class XII (Karnataka School Examination and Assessment Board)	2023	98.67%
Jain Heritage School, Bangalore	Class X (Central Board of Secondary Education)	2021	97.80%

SCHOLASTIC ACHIEVEMENTS

Institute Academic Excellence Award (Freshman and Junior Year): Recognized for achieving the highest CGPA in the CSE department.

CSE Department Topper (Sophomore Year): Also achieved the highest CGPA in my batch during my sophomore year.

Kishore Vaigyanik Protsahana Yojana (KVPY) 2022: Qualified the prestigious exam conducted by IISc, Bangalore.

NTSE Scholar 2021: Cleared both stages of the examination and am receiving a monthly scholarship from the Government of India

Quadeye Scholarship Program: Selected for the program after successfully clearing the Quadeye Scholarship Test and interview process.

Reached the **final round** of **Quantitative Trading internship interviews** at **Jane Street** in India.

Competitive Programming: I am currently an **expert** on **Codeforces.com,** and have a rating of **1627.** Here is my profile.

Teaching Assistant: I have been a teaching assistant for the courses **Discrete Mathematics, Data Structures** and **Algorithms**.

WORK EXPERIENCE

Software Engineering Intern: Databricks | Kafka Team (Under Storage Platform)

- Onboarded **Kafka onto Databricks' storage platform**, integrating Kafka with the existing control-plane framework through **Temporal and the regional Messaging Ecosystem Manager (MEM)** to enable centralized, durable, and automated operational workflows.
- Built the **Kafka auto-remediation engine and automated rebalance workflow** using **Scala, Temporal, Strimzi, and Cruise Control**, replacing manual partition rebalancing with health checks, rate limits, exclusive cluster locking, provisioning gates, approval, and post-execution verification. The auto-remediator was then extended to automate **disk-expansion workflows for Kafka brokers**.
- Implemented **daily metadata backfill** to automatically discover new Kafka clusters and regions, assign persistent UUIDs, and detect configuration drift, enabling true mutual exclusion across operations and automating **over 25 rebalances per capacity cycle**, which previously could take multiple weeks to complete and required constant manual monitoring.

PROJECTS

Built a Custom x86 Operating System

- Developing an **x86-based operating system** from scratch with support for keyboard input, **custom interrupts**, and a **Global and Interrupt Descriptor Table** setup. Built low-level components like **port handlers, interrupt handlers and keyboard drivers** using **C++ and Assembly**.
- Demonstrates proficiency in **systems and assembly programming**, **memory layout**, and **OS fundamentals** with modular code structure. The software can be run in a virtual environment like **QEMU or VirtualBox** too. The github link to the repository is provided here.

A Hybrid Routing Algorithm using Expander Graphs

- Designed a hybrid routing algorithm using **spectral graph clustering and expander-based hierarchical routing**, partitioning large networks into clusters and combining in-cluster paths with inter-cluster routing to reduce routing-table computation from traditional approaches.
- Achieved **14.6× faster setup than Link State, 4.9× faster than OSPF, and 11.9× faster than Distance Vector** on large networks, while maintaining **100% packet delivery** and comparable throughput and latency. Developed as a part of the Computer Networks course.

Lock-Free Parallel Algorithms for Implementing Fibonacci Heaps

- Implemented lock-free parallel Fibonacci heaps in C++20 using **immutable heap states, CAS-based synchronization, and concurrent memory management**, and developed a novel **persistent striped design** to reduce contention from a single global CAS bottleneck.
- Achieved up to **5.32× speedup** over the sequential implementation and improved scalability and throughput across multi-threaded workloads, reaching **1.089 Mops/s** on insert-heavy workloads. Developed as a part of the Parallel and Concurrent Programming course.

JADE: A Domain Specific Language for Distributed Computing

- Developed **JADE (Joint Asynchronous Distributed Environment)**, a DSL and compiler for distributed computing, with a complete pipeline from parsing and semantic analysis to C++/MPI code generation, including **topology-aware expander clustering and deadlock analysis**.
- I Implemented the optimization phase, including **spectral graph clustering and communication optimizations** using expansion and conductance metrics, alongside wait-for graph analysis to detect deadlocks. Developed as a part of the Compilers course.

Multilingual Manipulation-Aware Refusal Mechanism

- Developed a multilingual safety pipeline using **HuggingFace Transformers, PyTorch, IndicBERT, XLM-R, and MuRIL**, extending manipulation detection to English, Hindi, code-mixed, and Romanized Indic inputs through language preprocessing and transliteration.
- Preserved **deterministic, auditable refusal logic** while improving cross-lingual detection, achieving **0.815 bilingual recall and 0.7897 Hindi recall** through threshold tuning and calibration. Developed as a part of the Natural Language Processing course.

RISC-V Assembly Code Simulator:

- Developed a complete **RISC-V (RV64I) assembly simulator** with integrated cache functionality, assembly and complete debugging support. Built using **C++** with modular parsing, error checking, and execution control including **stepwise debugging** and **D-cache simulation**.
- Enabled accurate testing and visualization of **RISC-V assembly execution**, enhancing debugging and **system level cache understanding**. Since it was built as a part of the Computer Architecture course, the github link to the repository cannot be provided.

CCIRP: Central Communication and Intelligent Reminder Platform

- Developed a centralized multi-channel communication platform using **Next.js, TypeScript, FastAPI, MongoDB, Celery, Redis, and Kafka**, supporting email, SMS, and WhatsApp campaigns with scheduling, audience management, analytics, and AI-assisted content generation.
- Built asynchronous campaign dispatch, engagement tracking, dynamic recipient segmentation, and a **Gemini-powered assistant**, enabling centralized communication. It is currently being used across our campus. Developed as a part of the Software Engineering course.

SKILLS

Programming: C/C++/Python/Scala, Software and Systems Development, Database Management Systems: MySQL and PostgreSQL

Machine Learning and Relevant Frameworks (PyTorch, TensorFlow, OpenCV, scikitlearn, NumPy, Matplotlib, cvxpy)

Developer Tools: LaTeX, Git, GitHub, Markdown

RELEVANT COURSES

Systems: Operating Systems, Database Management Systems, Computer Architecture, Computer Networks, Compilers, Parallel and Concurrent Programming, Advanced Compiler Engineering, Advanced Computer Architecture, Software Engineering

Mathematics: Topics in Combinatorics, Probability and Random Variables, Linear Algebra, Transform Techniques, Statistics. Group Theory, Differential Equations, Calculus-I, Calculus-II, Linear Optimization

Theoretical Computer Science: Data Structures, Algorithms, Computational Number Theory and Algebra, Approximation Algorithms, Topics in Theoretical Computer Science, Discrete Mathematics

Machine Learning: Convex Optimization, Foundations of Machine Learning, Statistical Learning Theory, Deep Learning, Natural Language Processing, Fraud Analytics, Multi-Agent Systems, Reinforcement Learning